

NATURAL HISTORY AND PROGNOSIS

The granulomatous inflammation of sarcoidosis can remit spontaneously or with therapy, and therefore, the general prognosis of sarcoidosis is good. Pulmonary sarcoidosis resolves, improves, or stabilizes in 60 to 90 percent of patients even without treatment. Remissions often occur within the first 6 months after diagnosis, although it may take 2 to 5 years. The prognosis is generally favorable for liver and peripheral lymph node sarcoidosis as well. Skin lesions may resolve with or without scarring or pigmentary changes.

Almost all severe impairment from sarcoidosis results from the development of fibrosis, likely due to hyalinization of granulomatous inflammation. It remains unclear whether this fibrosis stems from inadequate treatment or an inherent host tendency toward a brisk fibrotic response. Cutaneous fibrosis can lead to permanent scarring and disfigurement. Rarely, cutaneous lesions may ulcerate or destroy adjacent bone or cartilage, particularly in patients with lupus pernio.

Because sarcoidosis is a systemic disease, prognosis is not solely determined by pulmonary involvement. In the United States, three-fourths of sarcoidosis-related deaths are due to pulmonary complications. Cardiac and central nervous system involvement account for most of the remaining deaths.

There is no current laboratory or radiologic test that can reliably predict disease outcome. Skin lesions are not a reliable prognostic indicator, but several associations have been identified:

- Patients with cutaneous lesions are more likely to have chronic systemic sarcoidosis compared to those without skin involvement.
- **Erythema nodosum**, especially when accompanied by fever and arthralgias, is associated with a favorable prognosis.
- Patients with **Löfgren syndrome** have more than an 80% rate of spontaneous remission, typically within 4 to 6 weeks.

Prognostic Factors in Sarcoidosis

Clinical Feature	Good Prognosis	Poor Prognosis
Erythema nodosum	+	
African descent		+
Extrathoracic disease		+
Stage I vs. II–III CXR	+	
Age > 40 years		+
Splenic involvement		+
Lupus pernio		+
Disease duration > 2 years		+
FVC < 1.5 L		+
Stage IV CXR / aspergilloma		+

CXR = chest x-ray; FVC = forced vital capacity